

Astroinformatics I: Practice 1

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1 Introduction

The goal of this practice was to download TESS light curves, convert FITS files into CSV format, manipulate the file lists using shell scripts, and visualize the light curves with TOPCAT.

2 Downloading the data

The light curve files from the TESS mission were downloaded from the Mikulski Archive for Space Telescopes (MAST) using the shell script also downloaded from STScI's Mikulski Archive.

The original shell script downloads many more light curves than requested, so it was necessary to terminate the process using **CTRL + C**.

This resulted in the following final files:

```
tess2023341045131-s0073-0000000001750268-0268-s_lc.fits
tess2023341045131-s0073-0000000001755406-0268-s_lc.fits
tess2023341045131-s0073-0000000001827744-0268-s_lc.fits
tess2023341045131-s0073-0000000001828620-0268-s_lc.fits
tess2023341045131-s0073-0000000001840666-0268-s_lc.fits
tess2023341045131-s0073-0000000001942416-0268-s_lc.fits
tess2023341045131-s0073-0000000001942969-0268-s_lc.fits
tess2023341045131-s0073-0000000001947463-0268-s_lc.fits
tess2023341045131-s0073-0000000001950736-0268-s_lc.fits
tess2023341045131-s0073-0000000001950893-0268-s_lc.fits
tess2023341045131-s0073-0000000002006984-0268-s_lc.fits
tess2023341045131-s0073-0000000002008765-0268-s_lc.fits
tess2023341045131-s0073-0000000002014191-0268-s_lc.fits
tess2023341045131-s0073-0000000002102329-0268-s_lc.fits
tess2023341045131-s0073-0000000002104696-0268-s_lc.fits
tess2023341045131-s0073-0000000002105589-0268-s_lc.fits
tess2023341045131-s0073-0000000002143575-0268-s_lc.fits
tess2023341045131-s0073-0000000002149979-0268-s_lc.fits
tess2023341045131-s0073-00000000053250605-0268-s_lc.fits
tess2023341045131-s0073-00000000055101449-0268-s_lc.fits
```

3 Fits to CSV

TOPCAT was used to read the FITS files; the same program was used to export the tables from the files in CSV format.

It should be noted that no error messages appeared when importing the FITS files.

4 Shell Scripts

4.1 Obtaining all the file names

For the first script, the ‘ls’ command was used to obtain the list of FITS files, and then the ‘>’ operator was used to write the output to a text file.

The resulting shell script is as follows:

```
#!/bin/bash

ls *.csv > csv_files.txt
```

4.2 Splitting the list

The split command in the first line divides the text file into multiple smaller files, each containing 5 lines. The resulted files are initially created without a file extension.

The second part of the script (the loop part) was rewritten specifically for compatibility, because of the differences between GNU and BSD `split` implementations, the original code was:

```
split -l 5 -d --additional-suffix=.txt csv_files.txt file_list_
```

this code could not be executed on macOS.

To solve the problem, a `for` loop was used to iterate through all files beginning with `file_list_` and rename them by adding the `.txt` extension.

The `for` loop solution was implemented to maintain compatibility with the GNU version but also to add compatibility to the BSD version on macOS.

The resulting shell script is as follows:

```
#!/bin/bash

split -l 5 -d csv_files.txt file_list_

for f in file_list_*; do
  mv "$f" "$f.txt"
done
```

5 Light Curve Plots

To make the plots, the software **TOPCAT** was used, to identify the correct *X* axis (time), only it was needed to read the columns until the one with the same name was found.

For the *Y* axis (luminosity) and with the same methodology, two possible columns were found, after plotting the graphs with both columns, it was determined that the value labeled **PDCSAP_FLUX** represents the normalized flow and will therefore be the one used.

After plotting all 20 light curves, it was possible to identify 3 different behaviors: well-defined and periodic, not well-defined, and well-defined but non-periodic light curves. All 20 plots can be described by one of these types. To avoid presenting all 20 plots, only the most representative case of each type will be presented here.

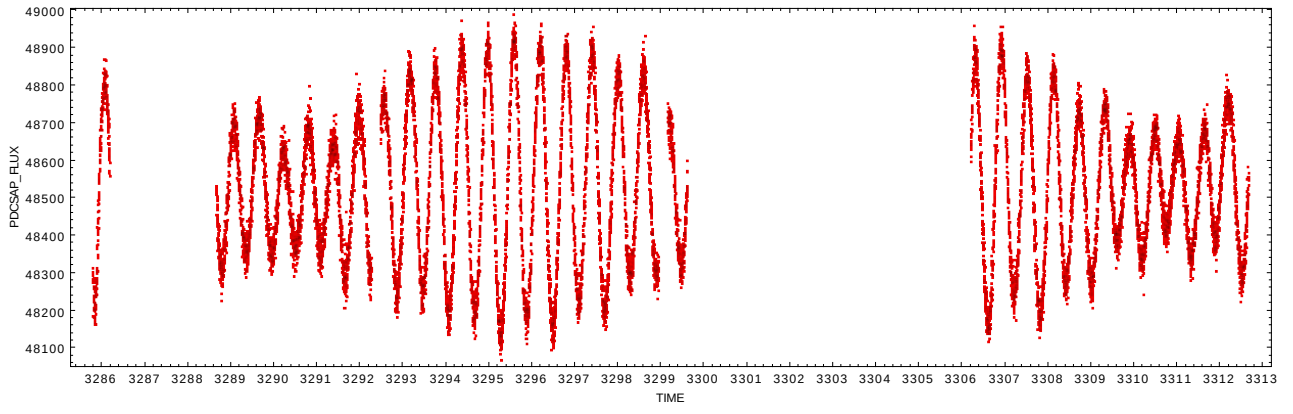


Figure 1: Well defined and periodical light curve.

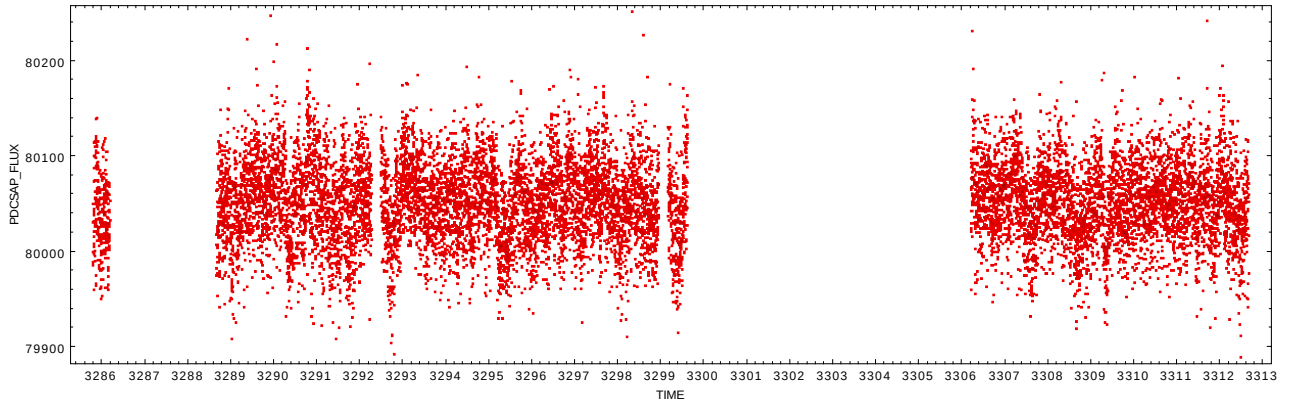


Figure 2: Non well defined light curve.

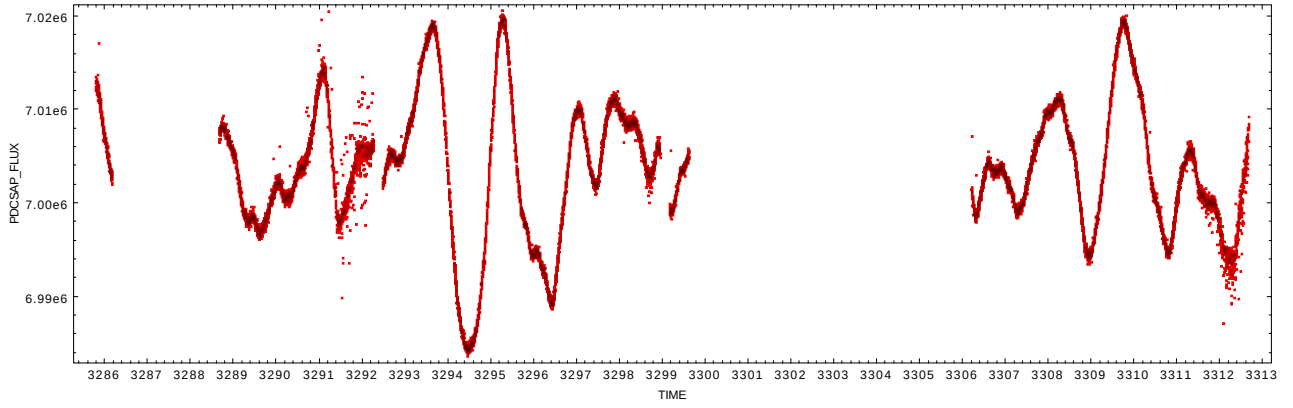


Figure 3: Well defined but non periodical light curve.